

Self-supervised Pre-training for Expert Finding

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Abstract

Expert finding plays a crucial role in Community Question Answering (CQA) platforms which aims to recommend questions to potential suitable users to answer. The key to expert finding is how to learn high-quality semantic representations for questions and experts. Most existing methods utilize limited supervised signals (e.g., expert-question interactions) to model experts/questions, and usually suffer from the data sparsity problem, which leads to sub-optimal performance. In this paper, to learn more comprehensive representations of questions and experts, we propose a Self-Supervised pre-training framework for Expert Finding (SSEF). The core is that we elaborately design two self-supervised pre-training tasks both toward question and expert learning respectively from large unsupervised data. Specifically, we propose an expert-level self-supervised module with two data-augmentation strategies including historical behavior crop or reorder. We pre-train each expert via contrastive learning from the augmented interaction question sequence that the expert has answered. Likewise, questions are pre-trained with another two data-augmentation strategies. Furthermore, the two self-supervised modules are unified by multi-task learning and share parameters of Pre-training Language Models (PLMs), which help to capture potential semantic relatedness between questions and experts. Extensive experimental results on six real-world CQA datasets demonstrate that our method achieves superior performance for expert finding.

1 Introduction

Recently, expert finding methods can effectively route questions to experts on Community Question Answering (CQA) websites, which has attracted considerable attention [Hu *et al.*, 2019; Yuan *et al.*, 2020b; Hu *et al.*, 2021; Li *et al.*, 2022; Peng *et al.*, 2022; Peng *et al.*, 2024]. Learning high-quality expert and question representations from historical interactions (i.e., questions answered by the expert historically) is the core of expert finding. Early methods [Zhou *et al.*, 2012;

San Pedro and Karatzoglou, 2014; Shahriari *et al.*, 2015] usually utilize handcrafted features or topic models to capture user and question characteristics. Recent methods [Li *et al.*, 2019; Ghasemi *et al.*, 2021; Peng *et al.*, 2022] usually learn question features from textual information and expert features from the historically answered questions. PMEF [Peng *et al.*, 2022] designed a multi-view paradigm (e.g., tag, title, body) to model experts and questions respectively, and then optimized the entire model with the supervised signal derived from expert-question interactions. These methods have achieved effective performance on expert finding.

Although these methods have achieved effective performance on expert finding. They rely on the observed historical interactions and usually suffer from the **data sparsity problem**. In other words, the observed interactions between experts and questions are often inadequate for modeling experts. For example, the densities of widely used *Es* and *Gis* datasets in StackExchange¹ are 0.0429% and 0.0435% respectively, which makes it very challenging to learn comprehensive representations of questions and experts. Recently, self-supervised pre-training, which learns general knowledge from large unlabeled data and alleviates the data sparsity problem, has been widely explored in various fields, like BERT [Devlin *et al.*, 2018] in Natural Language Processing. Inspired by this, in this paper, we aim to explore how to pre-train general expert and question representations from unlabeled data. However, it is nontrivial to conduct self-supervised pre-training in expert finding due to the following challenges:

(i) How to pre-train both experts and questions simultaneously? Existing PLMs (e.g., BERT) focus on the corpus or sentence granularity, which may be suitable for pre-training questions but not experts, which cannot simply be represented by a sentence, but need to be represented from their historical answering behaviors. However, pre-training both experts and questions is necessary for the expert finding task. Therefore, it is important to represent both questions and experts with suitable formats as inputs to PLMs so that PLMs can pre-train both experts and questions effectively. Unfortunately, this is an open question for PLMs.

(ii) How to design specific self-supervised pre-training tasks for expert finding? It is well known that a suitable pre-

¹<https://archive.org/details/stackexchange>

training task is key to capturing the general knowledge from a corpus. However, the existing PLM tasks (e.g., Masked Language Model) are usually token-level pre-training and incapable of capturing expert-level representations. Hence, it is necessary to design suitable pre-training tasks for pre-training experts from their historical answering behaviors. Meanwhile, we also need a proper question-level pre-training task for the benefit of question representation learning.

To this end, we propose an effective self-supervised pre-training language model for expert finding (named as **SSEF**). The core idea of SSEF is that we elaborately design two self-supervised pre-training tasks for both expert and question representation learning respectively from large unlabeled data to address the two aforementioned challenges.

For **Challenge-i**, we construct a specific pre-training corpus for experts and questions including (1) *question-level information*, which contains the title of each question (one question, one input line) from CQA websites; (2) *expert-level information*, which aggregates the historical questions answered by one expert as one sequence (one expert, one input line). We further design two special tokens ([*ER*] and [*QR*]) at the beginning of two sequences to help the PLMs identify whether the input is an expert or a question. In this way, the model is endowed with the ability to capture expert and question representations simultaneously. For **Challenge-ii**, one of the crucial modules in self-supervised pre-training is data augmentation. Hence, considering that the expert interest could be learned from the historical behaviors (i.e., the answered questions), we propose two expert-level data augmentation methods (namely behavior crop and reorder) to augment the original *expert-level information*. Then the model can be optimized with expert-level contrastive learning. In this way, our method can pre-train expert representations by distinguishing the preference difference between experts. For questions, we employ two data augmentation strategies on the *question-level information* (namely word crop and reorder) and pre-train question features by distinguishing the intrinsic difference between questions.

The expert-/question-level self-supervised modules are unified by multi-task learning and share the parameters. In the fine-tuning phase, we utilize the pre-trained weight to encode expert/question respectively, and then accomplish the downstream expert finding task via a fine-tuning way. In summary, the contributions of our method are:

(1) We design a self-supervised pre-training framework for expert finding, and this pre-training and fine-tuning paradigm could alleviate data sparsity. Our code, pre-trained model, and validation data are anonymously available ².

(2) The model pre-trains experts and questions simultaneously. We design new data-augmentation approaches for expert-level and question-level self-supervised learning.

(3) To verify the effectiveness of SSEF, we perform extensive experiments on six real-world CQA datasets. Experimental results demonstrate that our method can achieve better performance than existing baselines and validate the effectiveness of our approach.

(4) In zero-shot setting, our method still shows competitive results. Another observation is that our pre-training weight could further improve other methods' performance.

2 Related Works

2.1 Expert Finding

To find capable experts for providing satisfactory answers to questions on CQA websites, the expert finding has received much attention from both research and industry community [Zhu *et al.*, 2011; Liu *et al.*, 2015; Roy *et al.*, 2018; Yuan *et al.*, 2020b; Wang *et al.*, 2024; Amendola *et al.*, 2024; Etemadi *et al.*, 2024]. Previous studies about expert finding could be categorized into *traditional methods* and *deep learning-based methods*. In most traditional methods, feature-engineering [Cao *et al.*, 2012] or topic-modeling technology [Guo *et al.*, 2008; Riahi *et al.*, 2012; Wei *et al.*, 2016] is employed to model the questions and experts, then routed target questions to the suitable experts. Deep learning-based methods employed the neural network to model experts and questions, then matched experts and target questions [Li *et al.*, 2019; Fu *et al.*, 2020; Ghasemi *et al.*, 2021; Liu *et al.*, 2022b]. For example, TCQR [Zhang *et al.*, 2020] employed a temporal context-aware attention model to learn the answerer representation, which could model temporal dynamics by multi-resolution settings.

Different from existing methods of learning expert and question representations that usually rely on supervised data, we design a self-supervised pre-training paradigm, which could integrate self-supervised signals from large unlabeled data and pre-train expert and question representations.

2.2 Self-Supervised Learning

Self-Supervised Learning (SSL) [Liu *et al.*, 2020] have shown outstanding performance on various machine learning tasks, especially in Computer Vision [He *et al.*, 2020; Newell and Deng, 2020], Natural Language Processing [Devlin *et al.*, 2018], etc. For example, BERT [Devlin *et al.*, 2018] used masked language models to capture deep bidirectional representations. However, 1) they almost are designed on corpus-level or sentence-level for learning general language knowledge; 2) data augmentation tasks in these methods (e.g., token shuffle/masking) are almost token-level, which would be not suitable for modeling expert. Hence, designing appropriate data augmentation tasks for expert finding is necessary.

It is noted that ExpertPLM [Peng and Liu, 2022] and ExpertBert [Liu *et al.*, 2022b] are also expert finding models with self-supervised paradigms. However, they 1) only focus on expert modeling and omit the question modeling simultaneously; 2) usually mask words in the expert-level input information for expert interest modeling, which is not customized for expert finding and could not distinguish the inherent interest difference between experts.

3 PROBLEM STATEMENT

In this section, we formalize the expert finding. We represent column vectors and matrices by bold lower case letters

²<https://www.dropbox.com/scl/fo/h7rm3mqqtukzhjp7ihle/h?rlkey=nlxy0g0fx815xyq8fsxscekwy&dl=0>

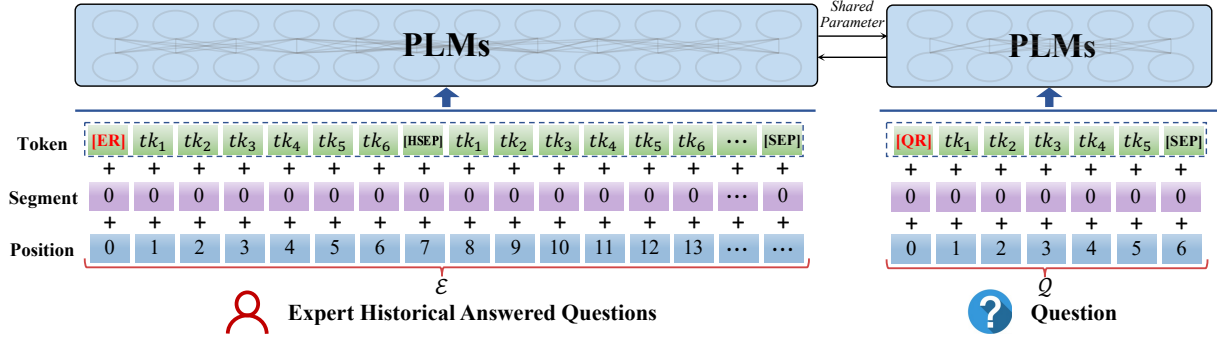


Figure 1: SSEF Pre-training Framework. We feed the expert-level information (e.g., $q_1, q_2, \dots$) and the question-level information (e.g., q) into BERT for pre-training expert/question. Furthermore, we add special tokens [ER] and [QR] at the begin of expert and question respectively, which could help the model identify whether the input is an expert or a question.

(e.g., e) and bold upper case letters (e.g., $\mathbf{X}$), respectively. Then, we formulate the problem definition of expert finding in our paper. Suppose that there is a target question q^t and a candidate expert set $\mathcal{C} = \{c_1, \dots, c_N\}$ respectively, where N is the number of experts. We need to compute the matching score s_c measuring the relevance between the expert in $\mathcal{C}$ and the target question q^t . Then, candidate experts are ranked and recommended to the target question q^t based on the s_c . Furthermore, the candidate expert $c_i \in \mathcal{C}$ is associated with a set of her/his historical answered questions, which can be denoted as $\mathcal{E}_i = \{q_1, \dots, q_m\}$, where m is the number of historical questions. Any question is represented by its related title, which is denoted as a word sequence $\mathcal{Q}$. Note that the expert who provides the “accepted answer” to the question will be regarded as the ground truth.

4 Proposed Method

In this section, we will introduce the training procedure of our method *SSEF*, which is composed of the following stages:

(1) In pre-training, the expert-/question-level pre-training inputs are first well constructed respectively. Then we propose expert-/question-level data augmentation methods to enhance original expert/question information and perform self-supervised tasks to fully model expert/question from unlabeled input data respectively. (2) In fine-tuning, we initialize the parameters of the expert or question representation model with the pre-trained parameters and then utilize traditional expert finding supervised signals to fine-tune the model.

4.1 Pre-training Framework

In this part, we will present the expert/question pre-training framework, including the well-designed input layer and a brief backbone architecture introduction.

For expert-level input, we concatenate the words of the expert’s historically answered questions into a whole sequence as shown in Figure 1. We add [HSEP] tokens between histories to distinguish different historical questions, and the special token [SEP] at the end of the input word sequence for recognizing the end of the sequence. Moreover, we propose to design a special token ([ER]) at the beginning of the input sequence, for indicating the expert information and help obtaining representations. Hence, the expert-level information

is denoted as:

$$\mathcal{E} = [[\text{ER}], \underbrace{[tk_1], [tk_2]}_{q_1}, [\text{HSEP}], \underbrace{[tk_1], [tk_2], [tk_3]}_{q_2}, [\text{SEP}]] \quad (1)$$

Likewise, we propose a special token [QR] at the beginning of the question-level information to facilitate obtaining the question representation, which is defined as follows:

$$\mathcal{Q} = [[\text{QR}], \underbrace{[tk_1], [tk_2], [tk_3], [tk_4], [tk_5]}_q, [\text{SEP}]] \quad (2)$$

The special tokens [ER] and [QR] are not just for capturing the representation of experts and questions. More importantly, they could help the model identify whether the input data is an expert or a question, which can be employed throughout the pre-training and fine-tuning.

Then, the input representation matrix $\mathbf{X}$ is constructed by summing its corresponding token, segment, and position embedding:

$$\mathbf{X} = \mathbf{X}_{\text{token}} + \mathbf{X}_{\text{seg}} + \mathbf{X}_{\text{pos}}, \quad (3)$$

It is noted that we don’t employ the Next Sentence Prediction task during pre-training, and hence, we set all the segment IDs to 0.

In our method, we adopt the widely used BERT [Devlin *et al.*, 2018] as our base backbone for further pre-training. For the detailed mathematical formulation of the model architecture (e.g., Self-Attention and Multi-Head mechanisms), please refer to Appendix A.

4.2 Data Augmentation

The appropriate Data Augmentation (DA) strategy plays a vital role in the self-supervised pre-training learning framework. In this part, we will show the well-designed data augmentation methods for experts and questions before the pre-training framework.

4.2.1 Expert-level Augmentation. For generating self-supervised signals to pre-train expert representation from historical questions, we design two augmentation approaches (shown in Figure 2 (a)(b)), which could construct different views of the same sequence but still maintain the main preference hidden in historical interactions.

(1) Behavior Reorder. In fact, we could obtain satisfied recommendation lists without strictly following the chronological order [Yuan *et al.*, 2019; Yuan *et al.*, 2020a]. Similarly, on CQA websites, it is likely that the expert answer

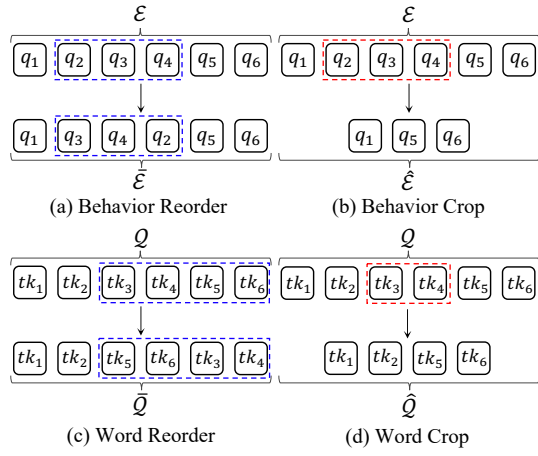


Figure 2: Data Augmentation Strategies.

questions in any order. For example, the sequence of historical questions is $[q_1]$ —“Single word for plaintiff or defendant”, q_2 —“Geometric or Geometrical?”, q_3 —“Replacement for this means that ...”. This is almost equivalent to $[q_2, q_1, q_3]$, since there is no strict order between questions.

Hence, we can derive a self-supervised augmentation operator with the assumption of flexible order. Specifically, given an expert historical question sequence $\mathcal{E}_i = \{q_1, \dots, q_m\}$, we randomly shuffle a continuous sub-sequence with augmentation ratio α_b (the length is $\alpha_b * |\mathcal{E}_i|$), which can be formulated as:

$$\bar{\mathcal{E}}_i = \text{reorder}(\mathcal{E}_i) \quad (4)$$

In this way, the SSEF could model the expert by relying less on the order of interaction sequences.

(2) Behavior Crop. In fact, randomly cropping a continuous sub-sequence in $\mathcal{E}$ might not affect the expert modeling greatly. This augmentation is designed to identify the influential historical questions for modeling expert, and make that less sensitive to the complete historical sequence. More concretely, given an expert historical question sequence $\mathcal{E}_i = \{q_1, \dots, q_m\}$, we randomly crop a continuous sub-sequence (with the length $\beta_b * |\mathcal{E}_i|$), which is denoted as:

$$\hat{\mathcal{E}}_i = \text{crop}(\mathcal{E}_i) \quad (5)$$

In this way, the SSEF could capture a local view of the expert histories and model more generalized expert representation without comprehensive information of experts.

4.2.2 Question-level Augmentation. We employ two augmentation approaches on the question-level information to construct different views of that. As shown in Figure 2 (c)(d), given the i -th question word sequence $\mathcal{Q}_i$, we employ **word reorder** [Yan *et al.*, 2021] or **word crop** augmentation strategies to change a continuous sub-sequence in one question, and the augmentation ratio is α_w or β_w (e.g., the changing length is $\alpha_w * |\mathcal{Q}_i|$). The operators are defined as follows:

$$\bar{\mathcal{Q}}_i = \text{reorder}(\mathcal{Q}_i), \hat{\mathcal{Q}}_i = \text{crop}(\mathcal{Q}_i). \quad (6)$$

It is noted that we only shuffle position IDs while keeping the order of the token IDs in the word reorder. This procedure encourages the model to identify the unique information that is shared across different views of the same question input, which helps learn more generalized question embeddings.

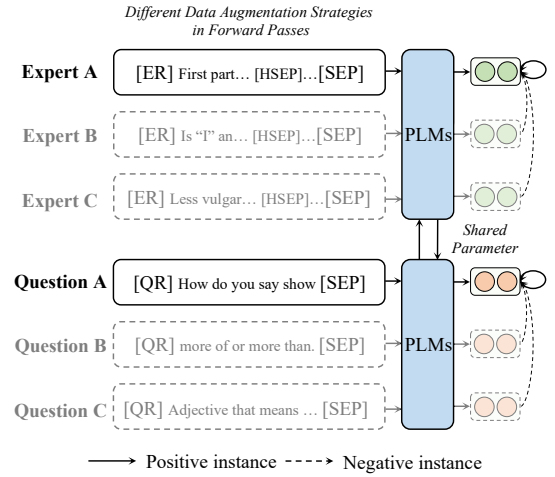


Figure 3: SSEF predicts expert/question itself from in-batch negatives. Note that expert or question self-supervised learning is simultaneous.

4.3 Self-Supervised Pre-training task

Having established the augmented views of the expert or question, we design two self-supervised tasks for expert and question respectively during the pre-training stage.

Expert Contrastive Task. This task teaches the model to discriminate the representation of a particular expert from other experts. Given j -th expert sequence, we have obtained the expert representation $\mathbf{e}_j$ (generated from [ER]). As shown in Figure 3, the augmented ones (randomly selected from question crop or reorder) are considered as positives (with the representation as $\bar{\mathbf{e}}_j$ or $\hat{\mathbf{e}}_j$), while other in-batch (with B is batch size) experts are considered as negatives. The expert-level self-supervised loss can be formally presented as follows:

$$\mathcal{L}_{ec} = - \sum_{j=1}^B \log \frac{\exp(\text{sim}(\mathbf{e}_j \cdot \bar{\mathbf{e}}_j) / \tau)}{\sum_{j'=1}^B \exp(\text{sim}(\mathbf{e}_j \cdot \hat{\mathbf{e}}_{j'}) / \tau)}, \quad (7)$$

$$\text{sim}(\mathbf{a} \cdot \mathbf{b}) = \cos(\mathbf{a} \cdot \mathbf{b}) = \frac{\mathbf{a}^T \mathbf{b}}{\|\mathbf{a}\|_2 \cdot \|\mathbf{b}\|_2}, \quad (8)$$

where τ is a temperature parameter. Since batches are constructed randomly, the in-batch negative augmented instances $\hat{\mathbf{e}}_{j'}$ would contain some experts from a mixture of multiple domains.

Question Contrastive Task. The object is to discriminate the representation of a particular question from other questions, which is similar to the expert contrastive task. Given i -th question, the question-level self-supervised loss can be denoted as:

$$\mathcal{L}_{qc} = - \sum_{i=1}^B \log \frac{\exp(\text{sim}(\mathbf{q}_i \cdot \hat{\mathbf{q}}_i) / \tau)}{\sum_{i'=1}^B \exp(\text{sim}(\mathbf{q}_i \cdot \bar{\mathbf{q}}_{i'}) / \tau)}, \quad (9)$$

where $\hat{\mathbf{q}}_i$ is i -th question augmented representation and $\bar{\mathbf{q}}_{i'}$ is the negative augmented instance.

Furthermore, for helping the model capture the CQA language knowledge, inspired by BERT [Devlin *et al.*, 2018], we simply mask 15% of the input tokens randomly (default

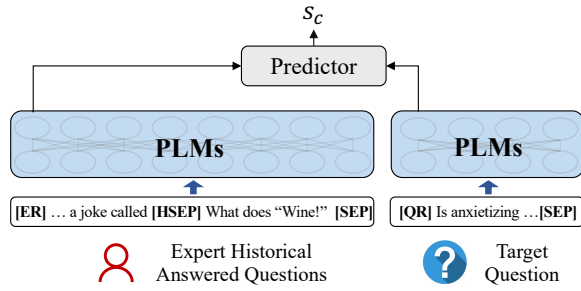


Figure 4: SSEF Fine-tuning. We employ the expert and question representations generated by [ER] and [QR] respectively for the downstream task.

in the BERT) and then predict those masked tokens. The formal definition is:

$$\mathcal{L}_{wm} = - \sum_{i=1}^{\mathcal{N}} \log p(n = n_i | \theta), n_i \in [1, 2, \dots, |\mathcal{V}|], \quad (10)$$

where $\mathcal{N}$ is the masked token set, $\mathcal{V}$ is the vocabulary.

Model Learning. At the pre-training stage, we leverage a multi-task training strategy to jointly pre-train the model:

$$\mathcal{L}_{pt} = \mathcal{L}_{ec} + \mathcal{L}_{qc} + \mathcal{L}_{wm}. \quad (11)$$

4.4 Fine-tuning for Expert finding

We have obtained the pre-trained representation model based on the unlabeled data. In this section, we will evaluate the expert finding task by fine-tuning the whole pre-trained model.

As illustrated in Figure 4, we initialize parameters of expert and question encoders with the pre-trained parameters. Afterward, we enter the expert historical answered questions and the target question into two encoders separately. Then, we concatenate the representations derived by [ER] and [QR] tokens for predicting the final matching score s_c . We employ negative sampling technology [Huang *et al.*, 2013] and sample K negative instances. Then we use cross-entropy loss to fine-tune our model as follows:

$$s_c = \frac{\exp(s_c)}{\sum_{j=1}^{K+1} \exp(s_j)}, \mathcal{L}_{ft} = - \sum_{c=1}^{K+1} \hat{s}_c \log(s_c), \quad (12)$$

where $\hat{s}_c$ is the ground truth label and s_c is the normalized probability predicted by the model.

5 Experiments

5.1 Experimental Setup

We conduct experiments on six real-world datasets from StackExchange, i.e., *Biology*, *CodeGolf*, *Gis*, *Es*, *Electronics* and *CodeReview*. We compare SSEF with eight competitive baselines: (1) Traditional Methods: NeRank [Li *et al.*, 2019], UserEmb [Ghasemi *et al.*, 2021] and ENEF [Liu *et al.*, 2022a]; (2) PLM-based Methods: TCQR [Zhang *et al.*, 2020], RMRN [Fu *et al.*, 2020], PMEF [Peng *et al.*, 2022], ExpertBert [Liu *et al.*, 2022b], and ExpertPLM [Peng and Liu, 2022]. We evaluate performance using Mean Reciprocal Rank (MRR), Precision@ K ($K = 1, 3$), and NDCG@20. Detailed dataset statistics, baseline descriptions, and parameter settings are provided in Appendix B, C, D.

5.2 Performance Comparison

We compare SSEF with baseline methods on six CQA datasets. The results are reported in Table 1, and we can find:

For the baseline methods, some earlier methods (e.g., Doc2Vec, CNTN) obtain poorer results on almost datasets, the reason may be that they usually employ max or mean operation on histories to model expert, which omits different history importance. NeRank utilizes the heterogeneous network embedding to unify raisers, answerers, and questions to learn embedding, which has obtained superior performance. Furthermore, the recent method PMEF has obtained superior performance compared with other baselines. One possible reason is that PMEF models the expert and question under a multi-view paradigm, which could be more comprehensive.

Compared with most baselines, SSEF performs consistently better than them on six datasets, which could verify the effectiveness of our proposed method. Different from those methods, we carefully construct expert-/question-level input information and corresponding training tasks, which could model expert/question generally in a self-supervised way and alleviate the supervised signal sparse issue. It is noted that the very recent works ExpertBert [Liu *et al.*, 2022b] and ExpertPLM [Peng and Liu, 2022] also achieve superior performance. This reflects the effectiveness of pre-training in expert finding. Our method outperforms them. The reason may be that they are preliminary pre-training models and omit the fact that supervised data is sparse in expert finding, while SSEF carefully designs data augmentation strategies and training tasks for self-supervised pre-training, which could learn more precise representations of experts/questions.

5.3 Zero-shot Performance

Considering our method is a pre-training method, we test the model performance with zero-shot setting. We only fine-tune the pre-trained model on the smallest dataset Biology, and test the performance on the Es and Electronics. The experimental results are shown in Table 3. It is surprising that the model can still achieve good performance (especially in P@3) by simply transferring from Biology to Es and Electronics without any fine-tuning. This could be attributed to the model has already captured a large amount of expert and question related knowledge during pre-training. And merely fine-tuning on smaller datasets (e.g., biology) is required to stimulate model understanding of the expert finding task, leading to zero-shot performance on other large datasets.

5.4 Ablation Study

Effect of Pre-training tasks

To highlight the effectiveness of our designing inputs and corresponding tasks, we design two model variants: **(a) Only QR.** We only employ the question-level information and corresponding question-level tasks for pre-training without expert-level pre-training; **(b) Only ER.** We only employ the expert-level information and corresponding expert-level tasks to pre-train without question-level pre-training; **(c) Original BERT.** We directly adopt the original BERT weight and training tasks for further pre-training based on the CQA corpus (one question, one input line), then fine-tuning on the expert

Table 1: Expert finding results of different methods. The best performance of the baselines is underlined. We perform t-tests and the results show that **SSEF** outperforms other baselines at significance level $p\text{-value} < 0.05$.

Dataset	CodeGolf				Gis				CodeReview			
Metric \ Method	MRR	P@1	P@3	NDCG@20	MRR	P@1	P@3	NDCG@20	MRR	P@1	P@3	NDCG@20
NeRank	0.4496	0.2616	0.5143	0.5341	0.4697	0.3032	0.5577	0.5836	0.4947	0.3089	0.6055	0.6154
TCQR	0.3225	0.2027	0.5087	0.5322	0.4553	0.2634	0.5489	0.5637	0.4253	0.2112	0.5382	0.5839
RMRN	0.4526	0.2321	0.5052	0.5375	0.4897	0.3239	0.5777	0.5832	0.4311	0.2517	0.5580	0.5892
UserEmb	0.3073	0.1857	0.4132	0.4530	0.3223	0.2433	0.4477	0.4562	0.3915	0.2031	0.5126	0.5246
PMEF	0.4232	0.2617	0.5066	0.5343	0.4861	0.3311	0.5888	0.5911	0.5020	0.3139	0.6161	0.6170
ENEF	0.3159	0.1927	0.3073	0.4326	0.4936	0.3337	0.5732	0.6023	0.4906	0.3103	0.6012	0.6120
ExpertBert	0.4431	<u>0.3032</u>	0.5195	<u>0.5816</u>	<u>0.5170</u>	<u>0.3537</u>	<u>0.6066</u>	<u>0.6265</u>	<u>0.5123</u>	<u>0.3287</u>	0.6265	<u>0.6248</u>
ExpertPLM	<u>0.4533</u>	0.3013	<u>0.5196</u>	0.5815	0.5085	0.3456	0.6058	0.6178	0.5115	0.3263	<u>0.6313</u>	0.6242
SSEF	0.4725	0.3235	0.5423	0.5912	0.5225	0.3676	0.6118	0.6275	0.5225	0.3399	0.6333	0.6399

Dataset	Es				Biology				Electronics			
Metric \ Method	MRR	P@1	P@3	NDCG@20	MRR	P@1	P@3	NDCG@20	MRR	P@1	P@3	NDCG@20
NeRank	0.5647	0.3932	0.6413	0.6617	0.4371	0.2886	0.5361	0.4991	0.5105	0.3459	0.5976	0.6221
TCQR	0.4716	0.3059	0.5941	0.6053	0.3962	0.2406	0.4422	0.4716	0.5016	0.3233	0.5953	0.6004
RMRN	0.5516	0.3761	0.6304	0.6453	0.4662	0.3153	0.5562	0.5578	0.5622	0.4038	0.6759	0.6729
UserEmb	0.4703	0.3032	0.5735	0.4869	0.3497	0.2263	0.3675	0.4672	0.4175	0.2954	0.4541	0.5170
PMEF	0.5735	0.4269	0.6520	0.6661	0.4719	0.3216	0.5693	0.5688	0.5872	0.4238	0.6950	0.6829
ENEF	0.5302	0.3732	0.6479	0.6338	0.4641	0.3138	0.5341	0.5593	0.5523	0.4196	0.6823	0.6911
ExpertBert	0.5726	0.4266	0.6513	0.6566	0.4859	0.3262	0.5701	0.5759	0.5923	0.4301	0.6958	0.6833
ExpertPLM	0.5840	<u>0.4380</u>	<u>0.6588</u>	<u>0.6793</u>	<u>0.4956</u>	<u>0.3353</u>	<u>0.5868</u>	<u>0.5929</u>	<u>0.6003</u>	<u>0.4356</u>	<u>0.7025</u>	<u>0.7046</u>
SSEF	0.5930	0.4433	0.6678	0.6893	0.5056	0.3466	0.5918	0.6109	0.6123	0.4526	0.7125	0.7099

Table 2: The variants of SSEF pre-training experiment results.

Dataset	Biology				CodeGolf				Electronics			
Metric \ Method	MRR	P@1	P@3	NDCG@20	MRR	P@1	P@3	NDCG@20	MRR	P@1	P@3	NDCG@20
Only QR	0.4831	0.3335	0.5701	0.5872	0.4468	0.2966	0.5235	0.5636	0.5995	0.4291	0.7003	0.6939
Only ER	0.4932	0.3389	0.5756	0.6036	0.4523	0.3015	0.5322	0.5801	0.6023	0.4321	0.7077	0.6958
Original BERT	0.4732	0.3281	0.5701	0.5772	0.4396	0.2887	0.5123	0.5532	0.5866	0.4201	0.6936	0.6818
SSEF	0.5056	0.3466	0.5918	0.6109	0.4725	0.3235	0.5423	0.5912	0.6123	0.4526	0.7125	0.7099

Table 3: Zero-shot Performance. Fine-tuning on Biology and testing on Es and Electronics.

Metric \ Dataset	MRR	P@1	P@3	NDCG@20
Es	0.5239	0.3335	0.6588	0.6344
Electronics	0.5602	0.3552	0.7012	0.6642

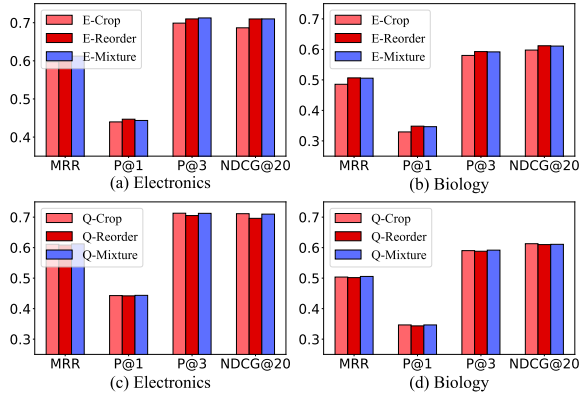


Figure 5: Performance comparison w.r.t various data augmentation strategies on experts or questions.

finding. As shown in Table 2, we can have the following observations: 1) **Original BERT** obtains the worst performance. This is equivalent to removing the question-/expert-level tasks from SSEF during pre-training. In this way, the model could lose the ability to model experts/questions. 2) It is surprising that the simple **Only QR** still outperforms the most of baselines and **Original BERT**. This is because **Only QR** could capture CQA language knowledge and benefit question-level representation learning during pre-training, which could help downstream expert finding tasks. 3) **Only ER** outperforms **Only QR**. This is because **Only ER** could pre-train expert-level representations that help model experts in downstream tasks. Furthermore, despite the absence of question-level sequences and tasks, **Only ER** allows the model to simply pre-train question representations due to the question sequences existing in the expert-level sequence. 4) Our complete model **SSEF** obtains the best results. The reason is that the **SSEF** could simultaneously pre-train the representations of expert/question with self-supervised tasks, which allows the model to capture unique representations for expert/question before performing the down-streaming task.

Effect of Different Data Augmentations

In this subsection, we turn to analyze the performance of SSEF with different augmentation strategies. **E-crop** or **E-Reorder** represents an enhanced view of the expert generated with only crop or reorder operation. **E-Mixture** repre-

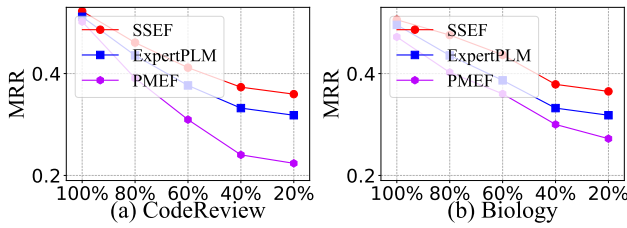


Figure 6: Performance comparison on SSEF and PMEF w.r.t. different amounts of training data.

sents randomly selecting one operation (crop or reorder) for generating the enhanced view. The experimental setting of the question side is similar. As shown in Figure 5, we can have the following observations: 1) The model effect exceeds the excellent baseline (PMEF) with different data enhancement strategies, which indicates that they could all assist in pre-training. 2) The composition of different augmentation strategies might not perform better than a single augmentation strategy. The reason may be that it is hard for the pre-training model to distinguish whether the enhanced data is transformed from the input data by different augmentation methods. 3) The **E-Reorder** usually performs better than **E-Crop**, while the question side is the opposite. This may be due to that performing the cropping operation directly may dramatically destroy the original expert behavior sequence and provide limited useful knowledge for downstream tasks. Finally, we employ the mixture data augmentation strategy to construct different augmented views.

5.5 Benefit from Pre-training

In this section, we further conduct detailed experiments to analyze the effect of our self-supervised pre-training model. We select a competitive method PMEF as a baseline in the following aspects.

Effect of Training Data Ratio in Fine-tuning

In this part, we investigate whether SSEF can alleviate the data sparsity problem. We simulate the data sparsity scenario by using different proportions of the full training data (CodeReview and Biology datasets as examples), including [100%, 80%, 60%, 40%, 20%]. Meanwhile, the ratios of validation and testing data remain the same (i.e., 10%) with the previous experiments.

As shown in Figure 6, we can find that there are growing gaps between the results of SSEF and PMEF with the reduction of training data. The reason may be that our model SSEF could learn more general representations for experts and questions during the pre-training stage, which could reduce the dependency on training data during the fine-tuning stage. In addition, we can find the gap in the CodeReview dataset is larger than that in the Biology dataset. This may be due to that the sparsity of the CodeReview dataset is more than the Biology dataset. In a word, experimental results imply that the SSEF shows potential to alleviate sparse supervised signal issue.

Applying the Pre-training Strategy on Other Models

Our designed pre-training strategy could learn more general expert and question representations, which have the potential to be applicable for supervised-based expert finding ap-

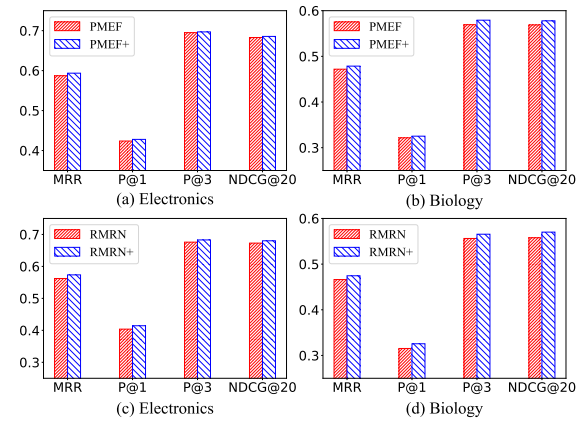


Figure 7: Performance comparison w.r.t. different base models enhanced by our proposed method SSEF.

proaches, e.g., PMEF [Peng *et al.*, 2022], RMRN [Fu *et al.*, 2020]. Therefore, we conduct experiments to explore whether the proposed pre-training strategy could also improve these recent expert finding approaches. We directly add the frozen pre-training expert/question representations to the PMEF and RMRN, named PMEF+ and RMRN+ respectively.

We report the results on the public dataset Electronics and Biology in Figure 7. From this figure, we can observe that our proposed method can consistently improve the performance of PMEF and RMRN, which further shows the effectiveness of the proposed method. In addition, we can find that the improvement effect of RMRN is better than that of PMEF. The reason may be that RMRN directly employs the original BERT word embedding weight, which has a semantic gap with the CQA domain.

5.6 Further Analysis

To verify the robustness of our model, we further investigate the impact of key hyperparameters on model performance, including the data augmentation ratios (varied in [0.1, 0.3]), the temperature parameter τ for contrastive learning (varied in [0.03, 0.5]), and the learning rate during fine-tuning (varied in [5e-08, 5e-04]). *The detailed experimental figures and comprehensive analysis are provided in Appendix F.*

6 Conclusion

In this paper, we propose SSEF, a self-supervised pre-training language model for the expert finding task on CQA. Different from existing supervised-based expert finding methods, the core of our method is to design a pre-training model for learn transferable expert and question representations in a self-supervised way. In particular, we proposed a model-agnostic framework SSEF to supplement the supervised expert finding task. We devise two types of data augmentation on question-level and expert-level information to construct auxiliary question/expert-level contrastive tasks, which could learn more universal expert/question representations for expert finding. We conduct detailed experiments on six real-world CQA datasets, and the results fully validate the effectiveness of our proposed pre-training method. Besides, another observation is that applying our pre-training weight to other methods could further improve their performance.

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